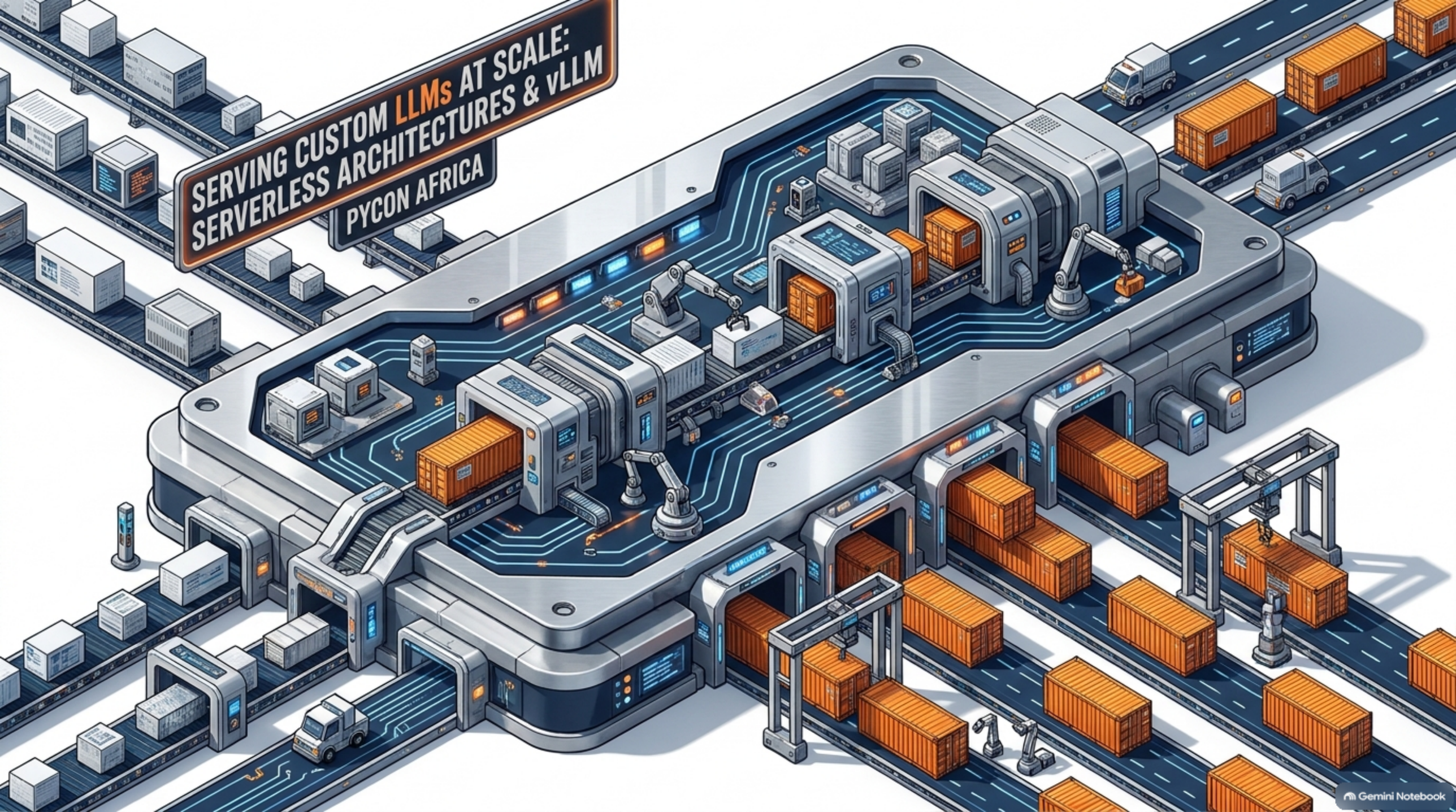
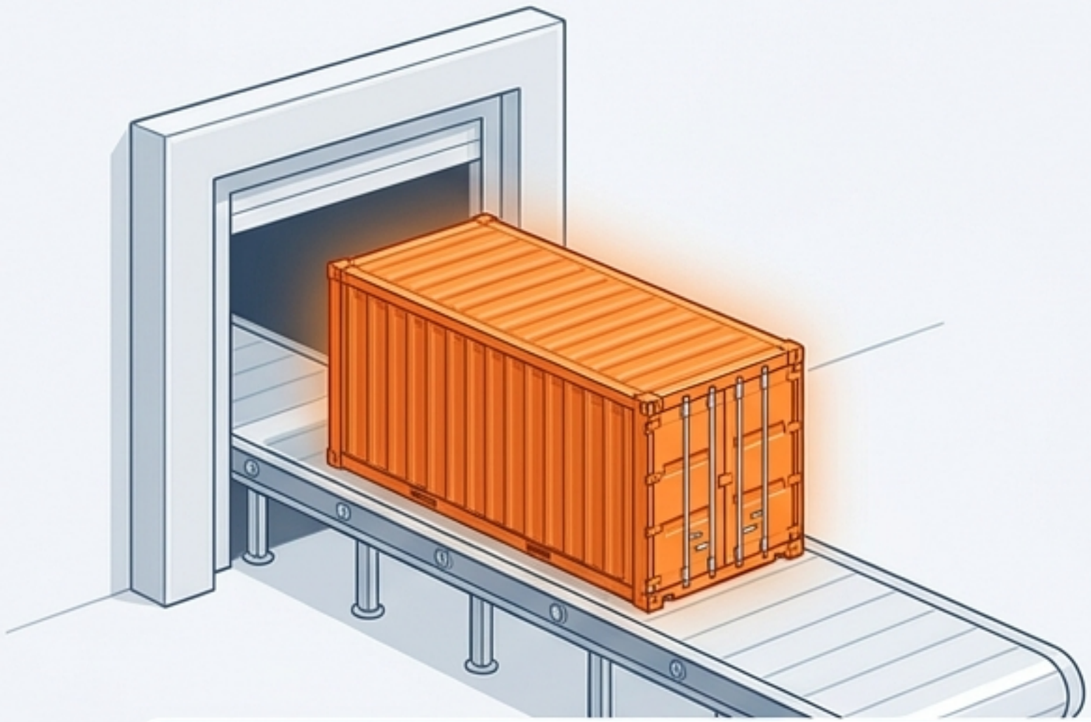

SERVING CUSTOM LLMs AT SCALE:
SERVERLESS ARCHITECTURES & vLLM
PYCON AFRICA
Gemini Notebook

The Dream

```
model.generate()
```



The Reality

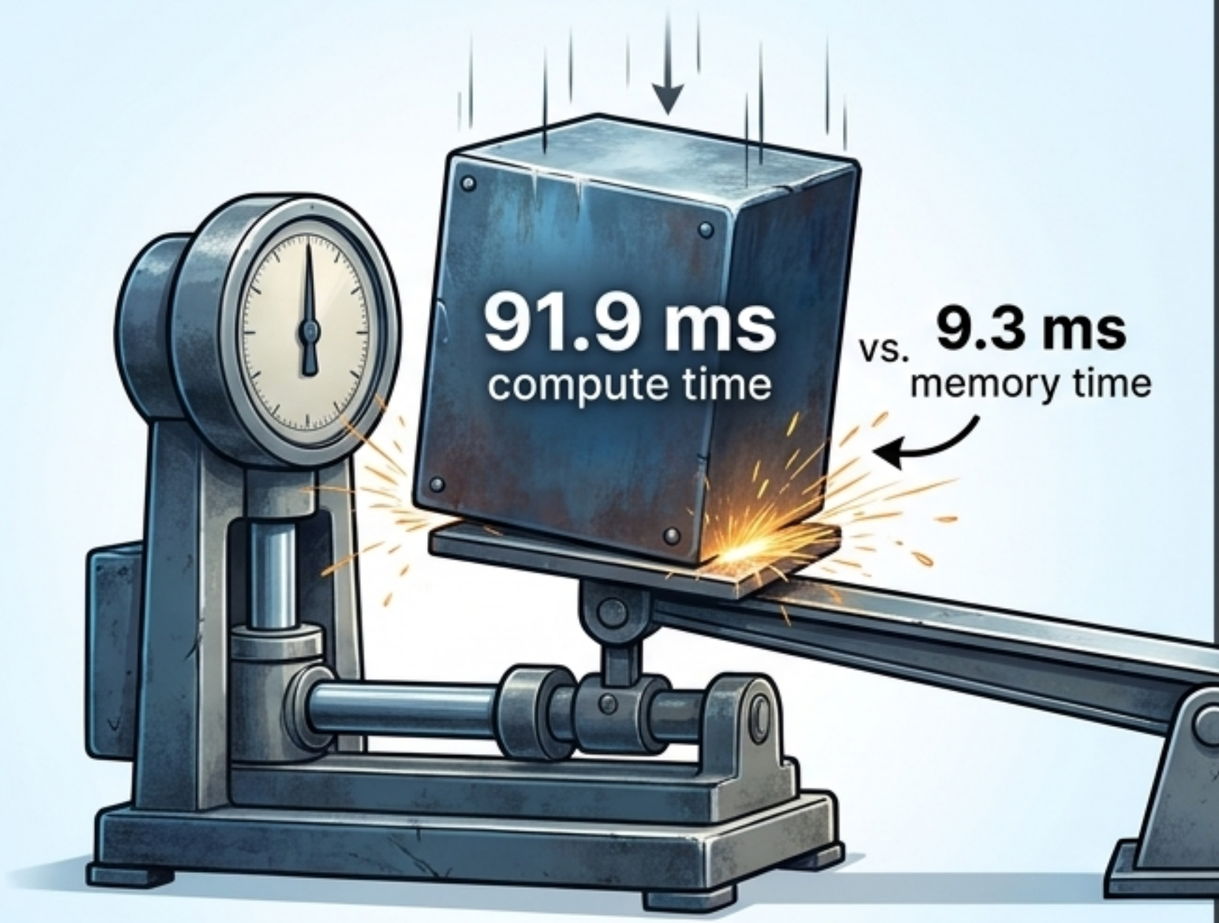
CUDA Out of Memory

"Standard libraries work perfectly for one user, but crash or crawl for 100 users. The problem isn't the model; it's the logistics of the memory."

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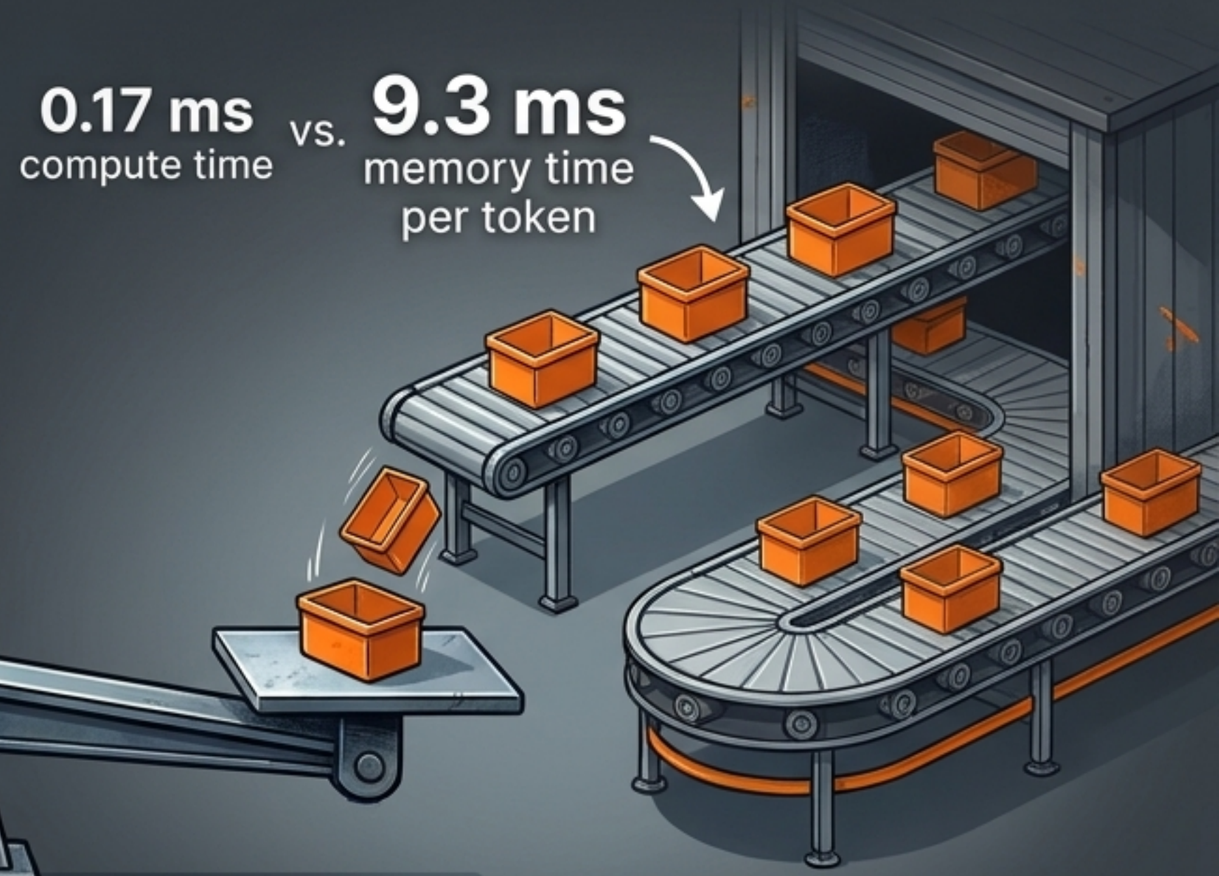
The Anatomy of an Answer: Prefill vs. Decode

Prefill Phase
Compute Bound



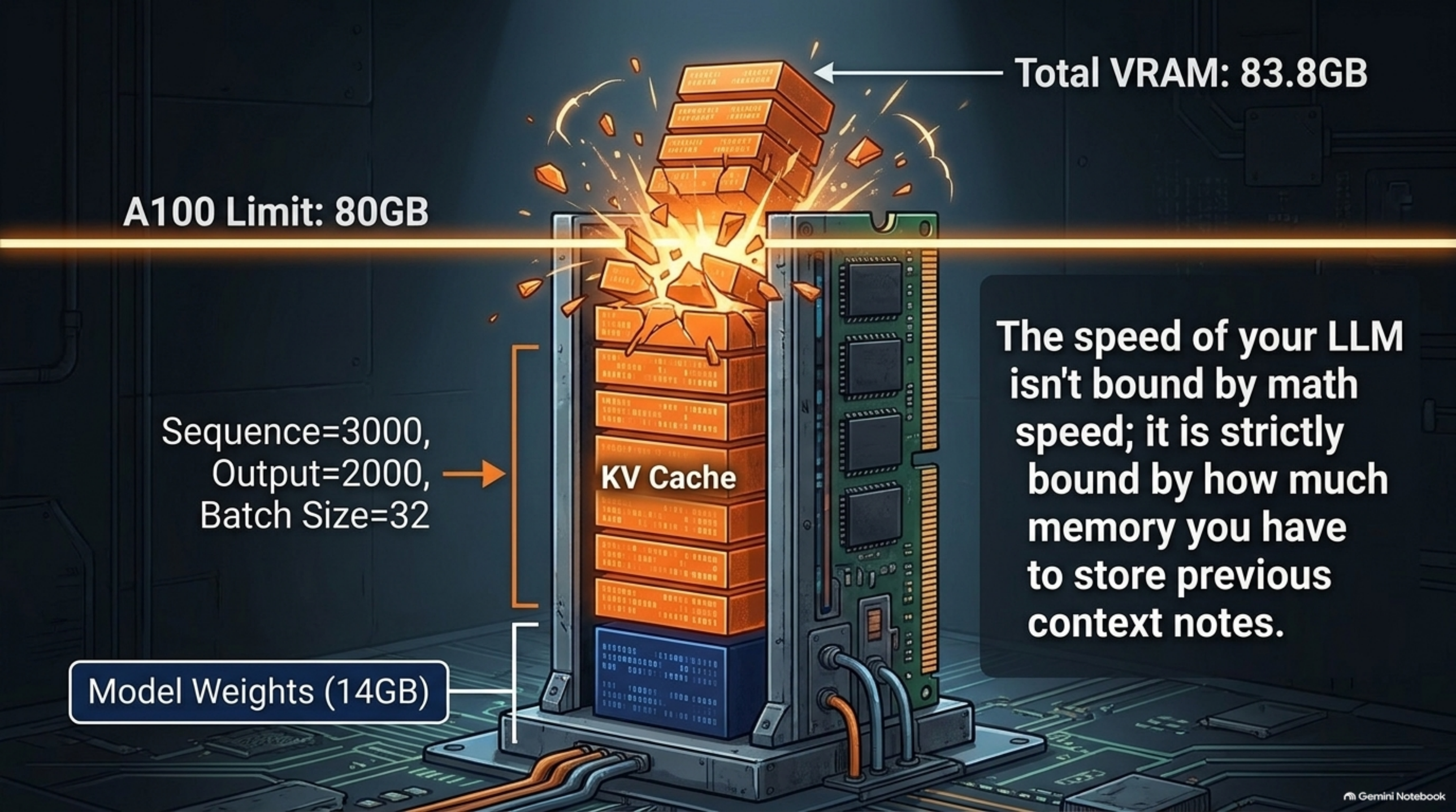
Model reads everything in one parallel forward pass.

Decode Phase
Memory Bound

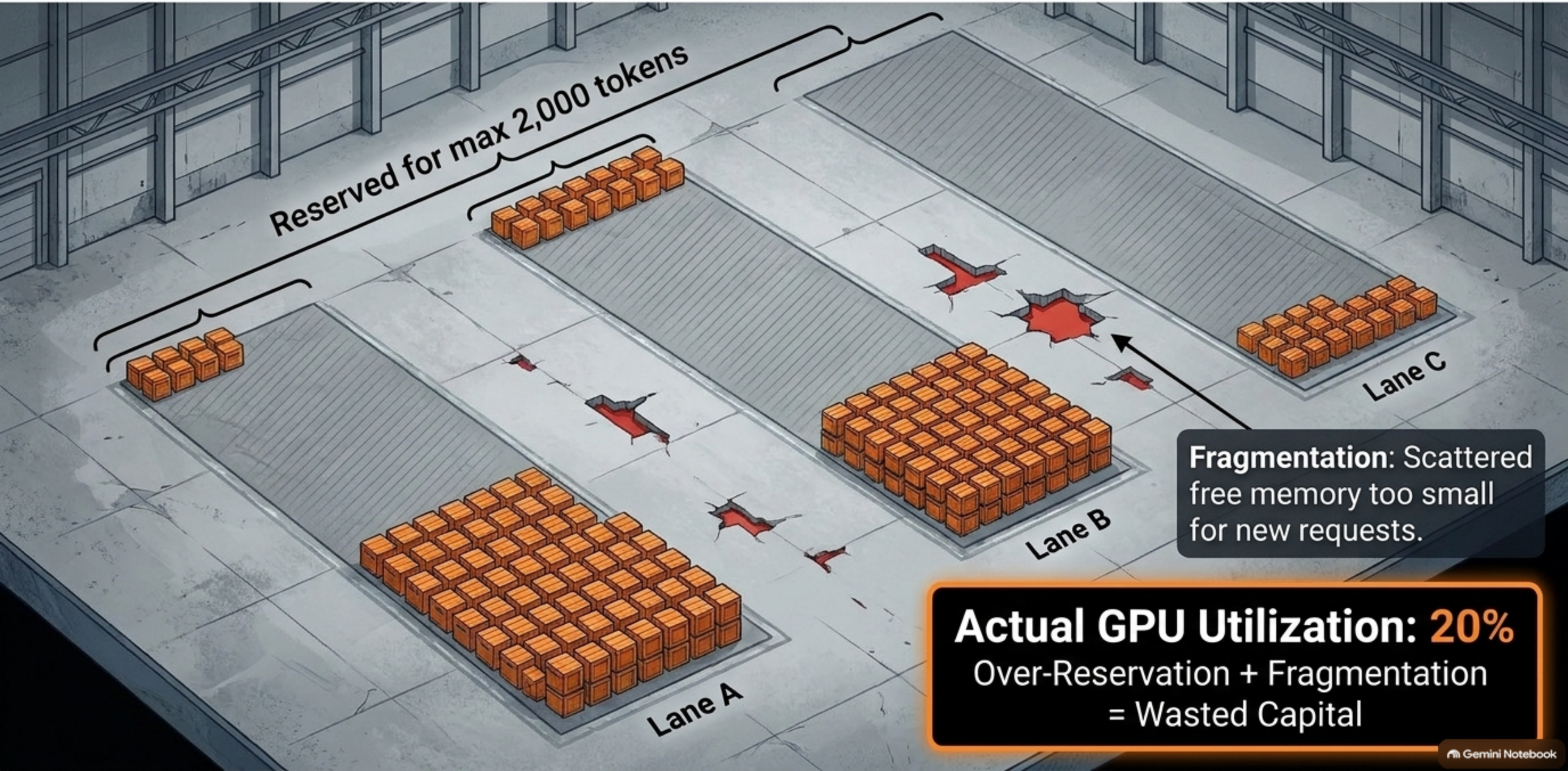


Model writes one word at a time, requiring constant memory retrieval.

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Total VRAM: 83.8GB
A100 Limit: 80GB
Sequence=3000,
Output=2000,
Batch Size=32
KV Cache
Model Weights (14GB)
The speed of your LLM isn't bound by math speed; it is strictly bound by how much memory you have to store previous context notes.
Gemini Notebook

Why Naive Serving Wastes GPU Memory and Money



Gemini Notebook

The vLLM Paradigm Shift: Treat the GPU Like an Operating System

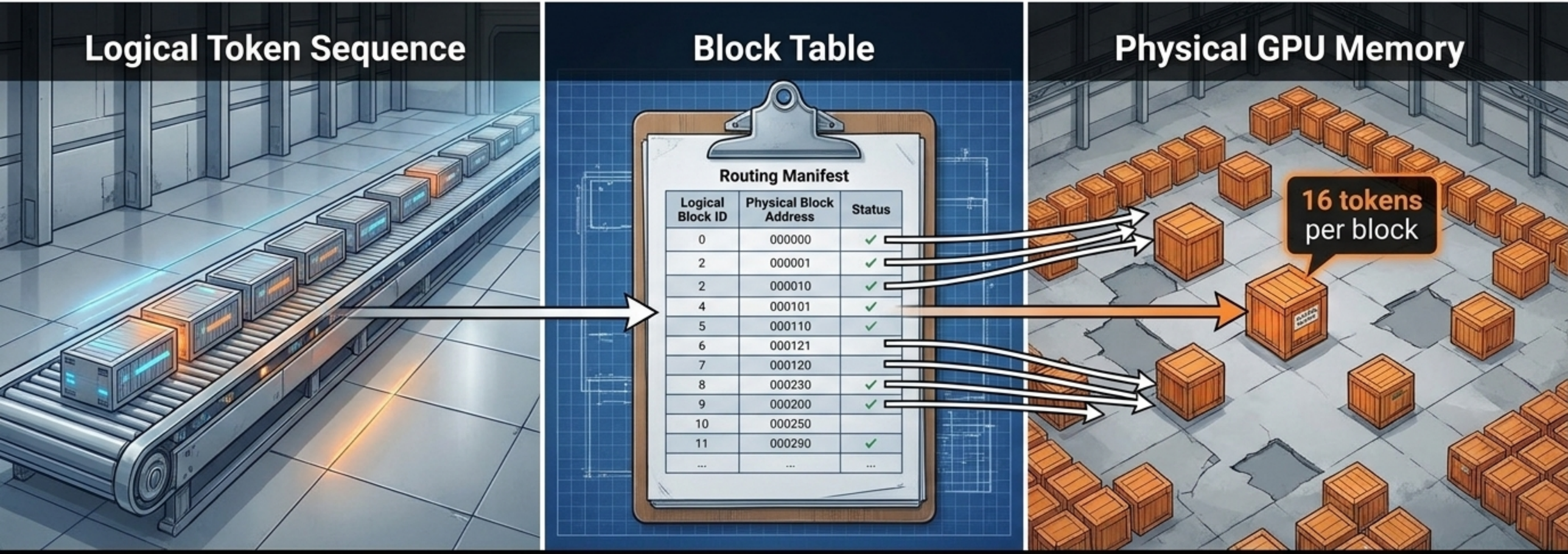


The memory problem was solved in the 1960s with OS Virtual Memory. vLLM brings this exact concept to the GPU.

Achieves up to 24x higher throughput than standard Hugging Face Transformers.

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PagedAttention: The Engine's Heart

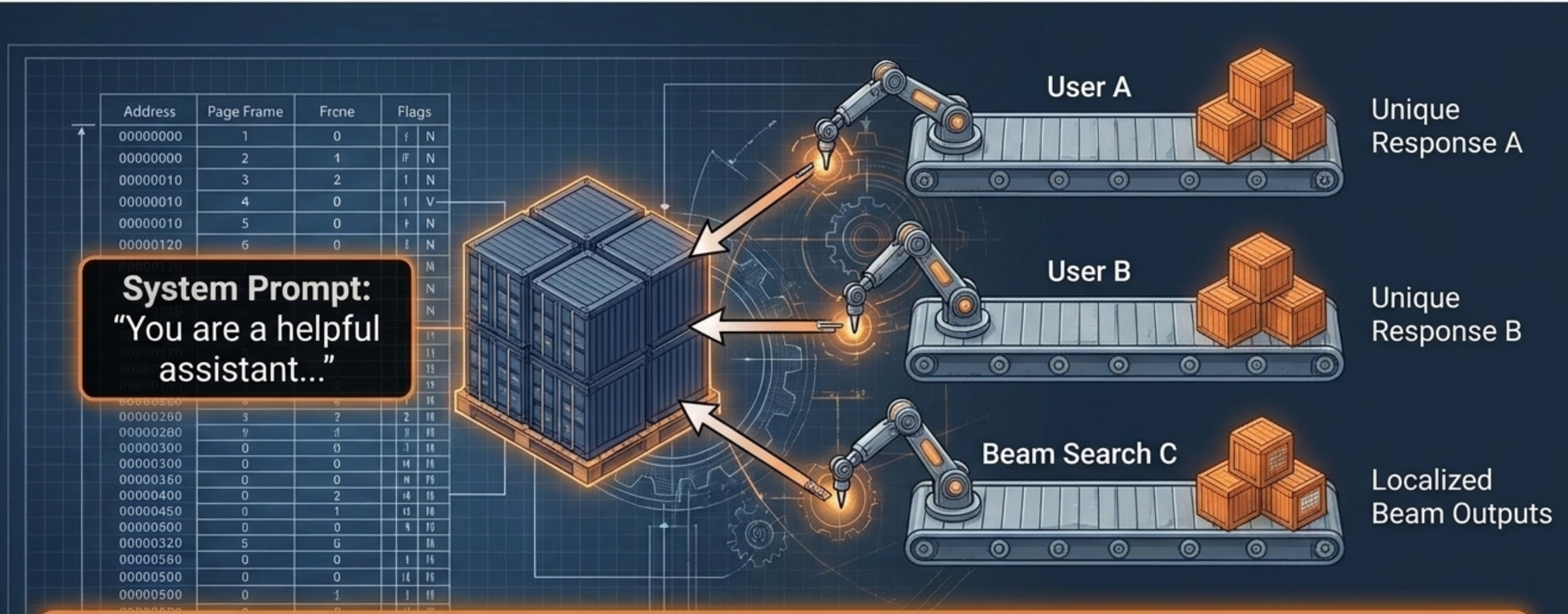


Zero Over-reservation:
Blocks allocated strictly on-demand.

Near-Zero Fragmentation:
Waste drops to under 4% because every block is uniform.

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The Free Superpower: Identical Prefix Sharing



Storing shared context once slashes **memory usage**, making vLLM the ideal engine for complex **Agentic** workflows, RAG, and Beam Search.

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Continuous Batching: Zero Idle GPU Time



Swap finished requests out and new ones in at every single token step.

Gemini Notebook

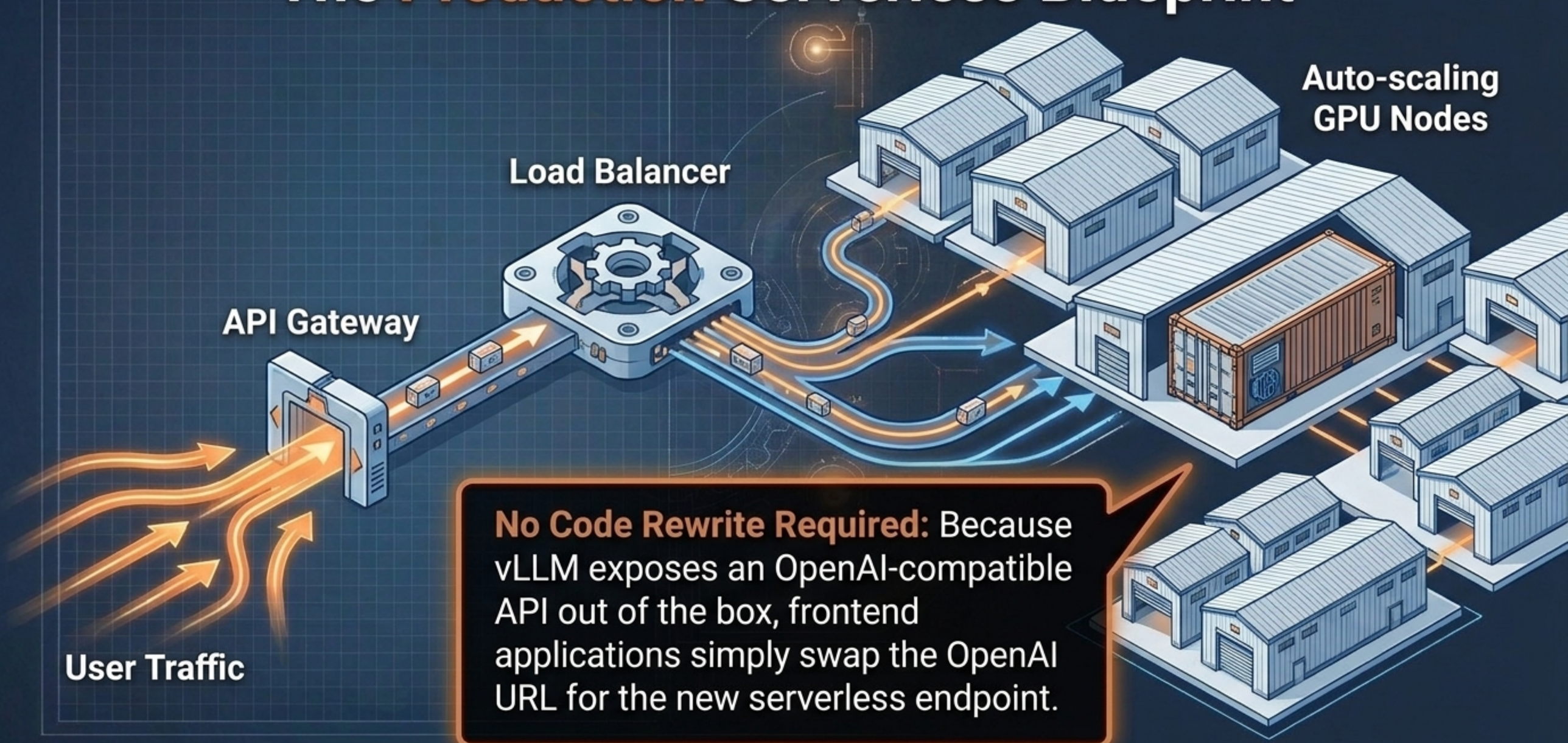
The Inference Engine Landscape

Framework	Core Optimization	Best Fit Use-Case	Primary Hardware Target
vLLM	PagedAttention & Continuous Batching	High-throughput general cloud serving	General GPUs (NVIDIA/AMD/AWS)
TGI	Rust router & Production Tracing	HuggingFace ecosystem & strict production	General GPUs
SGLang	Radix Trees & Speculative Decoding	Deep prefix sharing & Agentic Workflows	CUDA/ROCm/XPU
llama.cpp	GGUF & Quantization	Edge, Local, Mac development	CPU, Apple Silicon, Consumer GPUs

Choose your engine based on your **logistical constraints**—scale, hardware, or complexity.

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The Production Serverless Blueprint

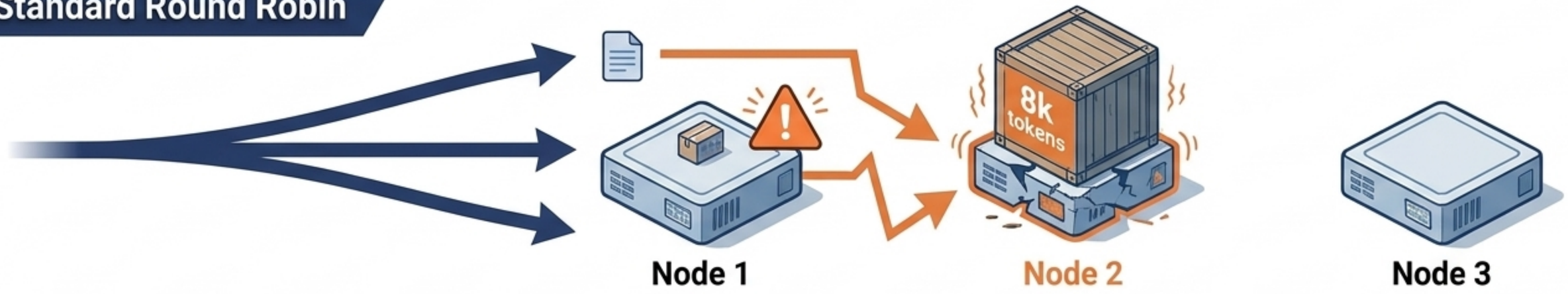


No Code Rewrite Required: Because vLLM exposes an OpenAI-compatible API out of the box, frontend applications simply swap the OpenAI URL for the new serverless endpoint.

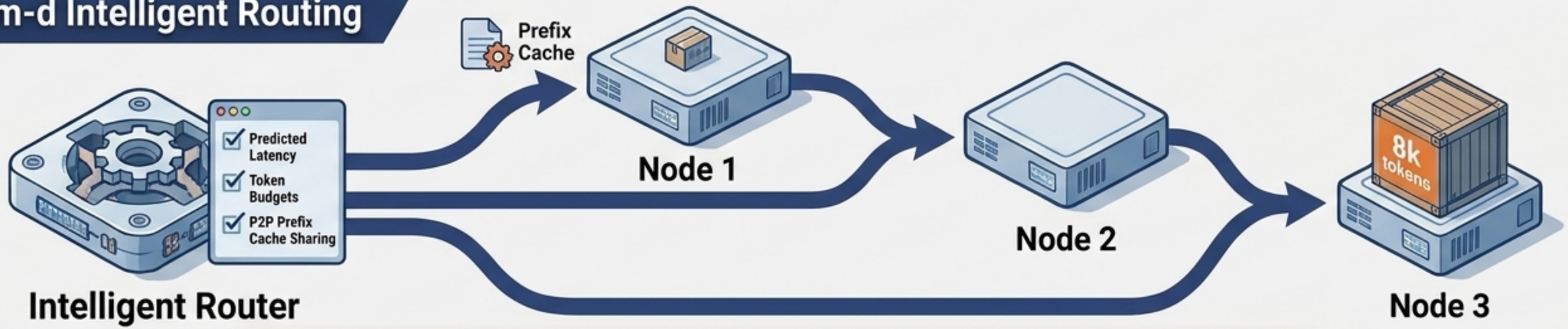
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Advanced Scaling: Intelligent Token Routing (llm-d)

Standard Round Robin



llm-d Intelligent Routing

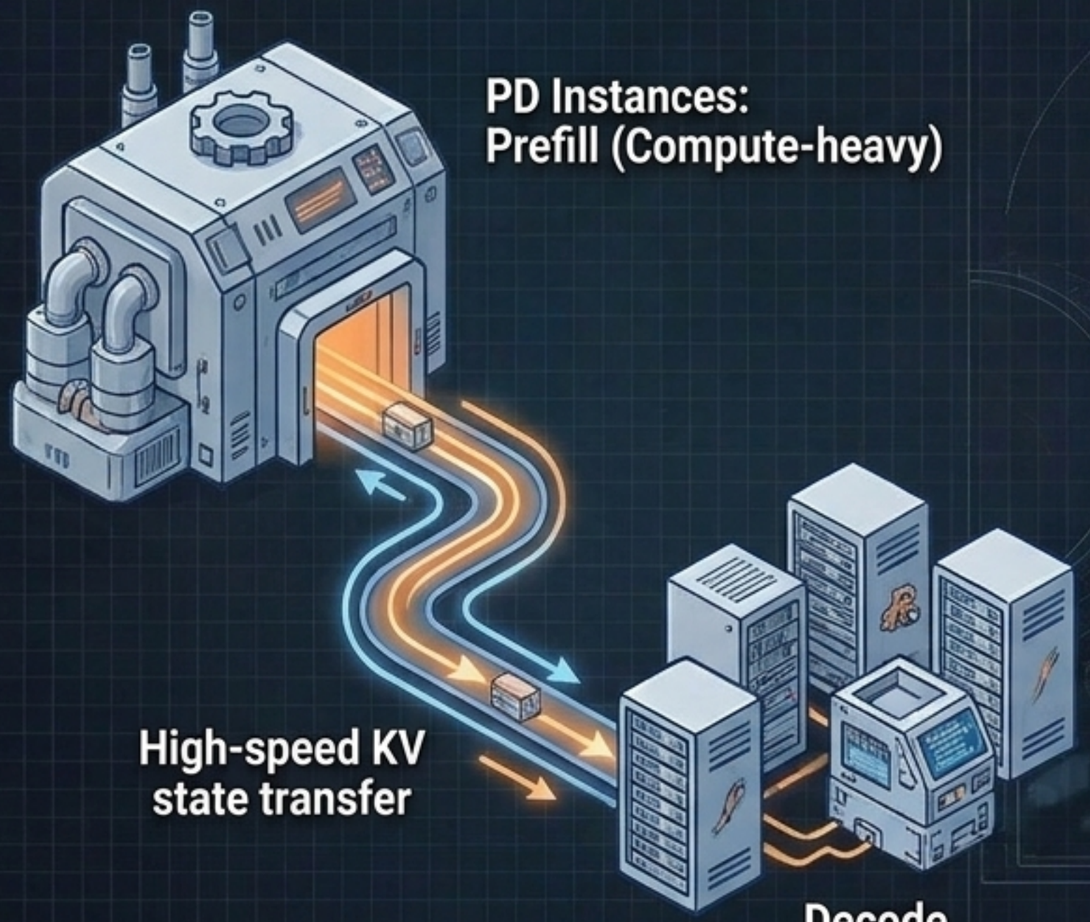


In a serverless LLM world, **simple load balancing fails**.
You must route based on real-time token budgets and KV cache state.

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Pushing the Limits: Disaggregation & Quantization

Prefill/Decode Disaggregation



Quantization



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The Ultimate Inference Stack

Control tower, dispatchers

Serverless Architecture:
Intelligent Routers (llm-d / TGI)
& OpenAI APIs

Active warehouse floor, robotic forklifts

Inference Engine:
vLLM (PagedAttention +
Continuous Batching)

Ground floor,
robust foundations

Hardware:
Bare Metal / Cloud GPUs
(NVIDIA, AMD, AWS Inferentia)



Don’t just throw bigger GPUs at the problem.
Engineer the memory, optimize the logistics, and scale seamlessly.

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